



Kalman filter framework for estimating parameters

- You know that xKF can estimate state of a dynamic system given known parameters and noisy measurements.
- It is also possible to use a (nonlinear) KF to estimate parameters of a model given a known state and noisy measurements.
 - It is not always possible to have known x_k , but it can be possible in a highly calibrated laboratory-type environment, designed to find parameter values that are later used in a production environment.
 - Note that a nonlinear KF is needed even if we wish to estimate parameters of a linear model since the parameters enter the model equations in a nonlinear way.
- This week, we first consider how to estimate system parameters if state is known.
- Then, we will consider how to estimate *both* state and parameters of a system simultaneously.



Parameter estimation: State equation

- We denote true parameters of a particular model by θ .
- We'll use xKF to estimate $\hat{\theta}_k$ much like we estimated $\hat{x}_k$.
 - Therefore, we require a model of the dynamics of the parameters.
- By assumption, parameters either don't change or change slowly, so we can model them as constant:

$$\theta_k = \theta_{k-1}.$$

- However, this doesn't work very well. The covariance of θ tends to collapse before the filter has converged on a good set of parameters. It is better to model the parameters as being able to change via some small perturbation:

$$\theta_k = \theta_{k-1} + r_{k-1}.$$

- The small white noise input r_k is fictitious, but models the slow drift in system parameter values plus infidelity of model structure.



Parameter estimation: Output equation

- Output equation required by xKF must be a measurable function of parameter values. We use:

$$d_k = \begin{bmatrix} f(x_{k-1}, u_{k-1}, \theta, e_k) \\ h(x_k, u_k, \theta, e_k) \end{bmatrix},$$

where $f(\cdot)$ and $h(\cdot)$ are the state and output equations of the system model being identified, and e_k models noises and modeling error.

- Note that the notation for $f(\cdot)$ and $h(\cdot)$ has been modified slightly to include the parameters θ in the model explicitly.
- Non-time-varying numeric values required by the model may be embedded within $f(\cdot)$ and $h(\cdot)$, and are not included in θ .
- The history of "measurements" is $\mathbb{D}_k = \{d_0, d_1, \dots, d_k\}$.
- Also, e_k may be a combination of w_k and v_k , but is considered distinct here.



SPKF for parameter estimation: 1a

- Using this nonlinear state-space model, can implement parameter estimation using any nonlinear KF.
- We will look at using SPKF first, then EKF in the next lesson.
- Define augmented θ^a that combines randomness of parameters, sensor noise.
- This augmented vector is used in the estimation process as described below.
- As always, we proceed by deriving the six steps of sequential probabilistic inference.

SPKF step 1a: Parameter prediction time update.

- The parameter prediction step is approximated as:

$$\hat{\theta}_k^{a,-} = \mathbb{E}[\theta_{k-1}^a + r_{k-1} \mid \mathbb{D}_{k-1}] = \hat{\theta}_{k-1}^{a,+}.$$

- This makes sense, since the parameters are assumed constant.



SPKF for parameter estimation: 1b

SPKF step 1b: Error covariance time update.

- Covariance prediction step is accomplished by first computing:

$$\begin{aligned} \tilde{\theta}_k^{a,-} &= \theta_k^a - \hat{\theta}_k^{a,-} = \theta_{k-1}^a + r_k - \hat{\theta}_{k-1}^{a,+} \\ &= \tilde{\theta}_{k-1}^{a,+} + r_k. \end{aligned}$$

- Then directly compute desired covariance:

$$\begin{aligned} \Sigma_{\tilde{\theta},k}^{a,-} &= \mathbb{E}[\tilde{\theta}_k^{a,-} (\tilde{\theta}_k^{a,-})^T] = \mathbb{E}[(\tilde{\theta}_{k-1}^{a,+} + r_k)(\tilde{\theta}_{k-1}^{a,+} + r_k)^T] \\ &= \Sigma_{\tilde{\theta},k-1}^{a,+} + \Sigma_{\tilde{r}}. \end{aligned}$$

- Time-updated covariance has additional uncertainty due to the fictitious noise “driving” the parameter values.



SPKF for parameter estimation: 1c

SPKF step 1c: Predict system output d_k .

- To predict system output, need sigma points describing output.
- This, in turn, requires a set of $p + 1$ sigma points describing $\theta_k^{a,-}$, which we will denote as $\mathcal{W}_k^{a,-}$:

$$\mathcal{W}_k^{a,-} = \left\{ \hat{\theta}_k^{a,-}, \hat{\theta}_k^{a,-} + \gamma \sqrt{\Sigma_{\tilde{\theta},k}^{a,-}}, \hat{\theta}_k^{a,-} - \gamma \sqrt{\Sigma_{\tilde{\theta},k}^{a,-}} \right\}.$$

- From the augmented sigma points, the $p + 1$ vectors comprising the parameters $\mathcal{W}_k^{\theta,-}$ and the $p + 1$ vectors comprising the modeling error $\mathcal{W}_k^e,-$ are extracted.



SPKF for parameter estimation: 1c (cont.)

- Output equation evaluated using all pairs of $\mathcal{W}_{k,i}^{\theta,-}$ and $\mathcal{W}_{k,i}^{e,-}$ (subscript i denotes that i th vector is being extracted from original set), yielding sigma points $\mathcal{D}_{k,i}$ for time step k :

$$\mathcal{D}_{k,i} = \begin{bmatrix} f(x_{k-1}, u_{k-1}, \mathcal{W}_{k,i}^{\theta,-}, \mathcal{W}_{k,i}^{e,-}) \\ h(x_k, u_k, \mathcal{W}_{k,i}^{\theta,-}, \mathcal{W}_{k,i}^{e,-}) \end{bmatrix}.$$

- Finally, output prediction is computed as:

$$\begin{aligned} \hat{d}_k^- &= \mathbb{E} \left[\begin{bmatrix} f(x_{k-1}, u_{k-1}, \theta, e_k) \\ h(x_k, u_k, \theta, e_k) \end{bmatrix} \middle| \mathbb{D}_{k-1} \right] \\ &\approx \sum_{i=0}^p \alpha_i^{(m)} \begin{bmatrix} f(x_{k-1}, u_{k-1}, \mathcal{W}_{k,i}^{\theta,-}, \mathcal{W}_{k,i}^{e,-}) \\ h(x_k, u_k, \mathcal{W}_{k,i}^{\theta,-}, \mathcal{W}_{k,i}^{e,-}) \end{bmatrix} = \sum_{i=0}^p \alpha_i^{(m)} \mathcal{D}_{k,i}. \end{aligned}$$



SPKF for parameter estimation: 2a

SPKF step 2a: Estimator gain matrix L_k^θ .

- To compute estimator gain matrix, must first compute required covariance matrices:

$$\begin{aligned} \Sigma_{\tilde{d},k} &= \sum_{i=0}^p \alpha_i^{(c)} (\mathcal{D}_{k,i} - \hat{d}_k) (\mathcal{D}_{k,i} - \hat{d}_k)^T \\ \Sigma_{\tilde{\theta}\tilde{d},k}^- &= \sum_{i=0}^p \alpha_i^{(c)} (\mathcal{W}_{k,i}^{\theta,-} - \hat{\theta}_k^{a,-}) (\mathcal{D}_{k,i} - \hat{d}_k)^T. \end{aligned}$$

- Then, we simply compute $L_k^\theta = \Sigma_{\tilde{\theta}\tilde{d},k}^- \Sigma_{\tilde{d},k}^{-1}$.



SPKF for parameter estimation: 2b–2c

SPKF step 2b: Parameter estimate measurement update.

- Now, compute *a posteriori* parameter estimate by updating *a priori* prediction using estimator gain and the output innovation $d_k - \hat{d}_k$:

$$\hat{\theta}_k^{a,+} = \hat{\theta}_k^{a,-} + L_k^\theta (d_k - \hat{d}_k).$$

SPKF step 2c: Error covariance measurement update.

- The final step is calculated directly from the optimal formulation:

$$\Sigma_{\tilde{\theta},k}^{a,+} = \Sigma_{\tilde{\theta},k}^{a,-} - L_k^\theta \Sigma_{\tilde{d},k} (L_k^\theta)^T.$$



Summary

- To use xKF for parameter identification, must first define a relevant nonlinear state-space model.
- You have now learned one method that can be used to do so.
- You have also learned how to derive the SPKF for parameter estimation.
 - Prediction steps similar to, but simpler than SPKF for state estimation.
 - Update steps nearly identical to SPKF for state estimation.
 - Appendix to this lesson lists all the steps.
- Next step is to derive EKF for parameter estimation.



Nonlinear SPKF for parameter estimation (1)

State-space model:

$$\begin{aligned}\theta_{k+1} &= \theta_k + r_k, \\ d_k &= \begin{bmatrix} f(x_{k-1}, u_{k-1}, \mathcal{W}_{k,i}^{\theta,-}, \mathcal{W}_{k,i}^{e,-}) \\ h(x_k, u_k, \mathcal{W}_{k,i}^{\theta,-}, \mathcal{W}_{k,i}^{e,-}) \end{bmatrix}\end{aligned}$$

where r_k and e_k are independent Gaussian noise processes with means zero and $\bar{e}$, respectively, and having covariance matrices $\Sigma_{\bar{r}}$ and $\Sigma_{\bar{e}}$, respectively

Definitions:

$$\theta_k^a = [\theta_k^T, e_k^T]^T, \quad \mathcal{W}_k^a = [(\mathcal{W}_k^\theta)^T, (\mathcal{W}_k^e)^T]^T, \quad p = 2 \times \dim(\theta_k^a)$$



Nonlinear SPKF for parameter estimation (2)

Initialization: For $k = 0$, set

$$\begin{aligned}\hat{\theta}_0^+ &= \mathbb{E}[\theta_0] \\ \hat{\theta}_0^{a,+} &= \mathbb{E}[\theta_0^a] = [(\hat{\theta}_0^+)^T, \bar{e}]^T & \Sigma_{\hat{\theta},0}^{a,+} &= \mathbb{E}[(\theta_0^a - \hat{\theta}_0^{a,+})(\theta_0^a - \hat{\theta}_0^{a,+})^T] \\ \Sigma_{\hat{\theta},0}^+ &= \mathbb{E}[(\theta_0 - \hat{\theta}_0^+)(\theta_0 - \hat{\theta}_0^+)^T] & &= \text{diag}(\Sigma_{\hat{\theta},0}^+, \Sigma_{\bar{e}})\end{aligned}$$

Computation: For $k = 1, 2, \dots$ compute:

$$\begin{aligned}\text{State estimate time update:} \quad \hat{\theta}_k^- &= \hat{\theta}_{k-1}^+ \\ \text{Error covariance time update:} \quad \Sigma_{\hat{\theta},k}^- &= \Sigma_{\hat{\theta},k-1}^+ + \Sigma_{\bar{r}} \\ \text{Output estimate:} \quad \mathcal{W}_k^{a,-} &= \left\{ \hat{\theta}_k^{a,-}, \hat{\theta}_k^{a,-} + \gamma \sqrt{\Sigma_{\hat{\theta},k}^{a,-}}, \hat{\theta}_k^{a,-} - \gamma \sqrt{\Sigma_{\hat{\theta},k}^{a,-}} \right\} \\ \mathcal{D}_{k,i} &= \begin{bmatrix} f(x_{k-1}, u_{k-1}, \mathcal{W}_{k,i}^{\theta,-}, \mathcal{W}_{k,i}^{e,-}) \\ h(x_k, u_k, \mathcal{W}_{k,i}^{\theta,-}, \mathcal{W}_{k,i}^{e,-}) \end{bmatrix}\end{aligned}$$



Nonlinear SPKF for parameter estimation (3)

Computation: For $k = 1, 2, \dots$ compute:

$$\begin{aligned}
 \text{Output estimate (cont.):} \quad & \hat{d}_k = \sum_{i=0}^p \alpha_i^{(m)} \mathcal{D}_{k,i} \\
 \text{Estimator gain matrix:} \quad & \Sigma_{\tilde{d},k} = \sum_{i=0}^p \alpha_i^{(c)} (\mathcal{D}_{k,i} - \hat{d}_k)(\mathcal{D}_{k,i} - \hat{d}_k)^T \\
 & \Sigma_{\tilde{\theta}\tilde{d},k}^- = \sum_{i=0}^p \alpha_i^{(c)} (\mathcal{W}_{k,i}^{\theta,-} - \hat{\theta}_k^-)(\mathcal{D}_{k,i} - \hat{d}_k)^T \\
 & L_k^\theta = \Sigma_{\tilde{\theta}\tilde{d},k}^- \Sigma_{\tilde{d},k}^{-1} \\
 \text{State estimate meas. update:} \quad & \hat{\theta}_k^+ = \hat{\theta}_k^- + L_k^\theta (d_k - \hat{d}_k) \\
 \text{Error covariance meas. update:} \quad & \Sigma_{\tilde{\theta},k}^+ = \Sigma_{\tilde{\theta},k}^- - L_k^\theta \Sigma_{\tilde{d},k} (L_k^\theta)^T
 \end{aligned}$$